

ME 557- Additive Manufacturing Project 1: Flying Turtle

In this project, Fused Filament Fabrication (FFF) technology was employed to fabricate the Flying Turtle mechanism. The objective was to successfully print and assemble all 14 components while comprehending and debugging the effects of print parameters on mechanical performance, surface finish, and quality.

The different components were printed with a 0.4 mm nozzle on a Creality Ender-3 with PLA material type, and Cura was used for slicing the components. The standard temperature settings were 202-205°C for the nozzle and the bed temperature varied, with 62°C for small and easy parts and 52°C for the bigger components. The print speed was 50 mm/s, and the layer height was 0.15 mm for the Gear II, and 0.2mm was suitable for the rest.

Easy components such as the frames, gears, and input parts were printed with no complications and just required proper adhesions and some hairspray to be added to the build plate. The medium components, such as the shell and the base, were also printed well with adding brim type adhesion. In contrast, the challenging parts lived up to the category they were named after. The parts, such as the crankshafts and the turtle base, were difficult and often needed some changes to be done to print them, since the joints would always get fused due to the bed heat, and the whole print would be a waste. To accurately print these parts, a raft-type adhesion was required since it kept the base of the parts safe from fusing with no need for any supports. The nozzle temperature of 202°C was helpful, but the crankshaft printed well with 52°C bed heat, and the turtle base printed with 55°C. The optimal infill density that worked well for the crankshaft was 30% and 40% for the turtle base.

Optimal Settings to Slice the Components:

Settings	Easy Components	Complex Components
Infill	40-50% (Lines)	30-40% (Lines)
Nozzle Temp	205°C	202°C
Bed Temp	62°C	52-55°C
Layer Height	0.2mm	0.2mm (expect 0.15 for Gear II)
Support	Tree and Touching Build plate	Not required
Adhesion	Brim/Raft	Raft (Most helpful)

In conclusion, every printed component was successfully assembled together. The movable parts were a little creaky, but they functioned accurately with no problem. This project improved my knowledge of FFF printing and how to use to printer's temperature, adhesion, and layer settings to print the components with good quality. The project also accurately addressed the different levels of difficulty in printing and gave me the experience on how to print parts catering to their complexity.



Fig 1: The Flying Turtle Print